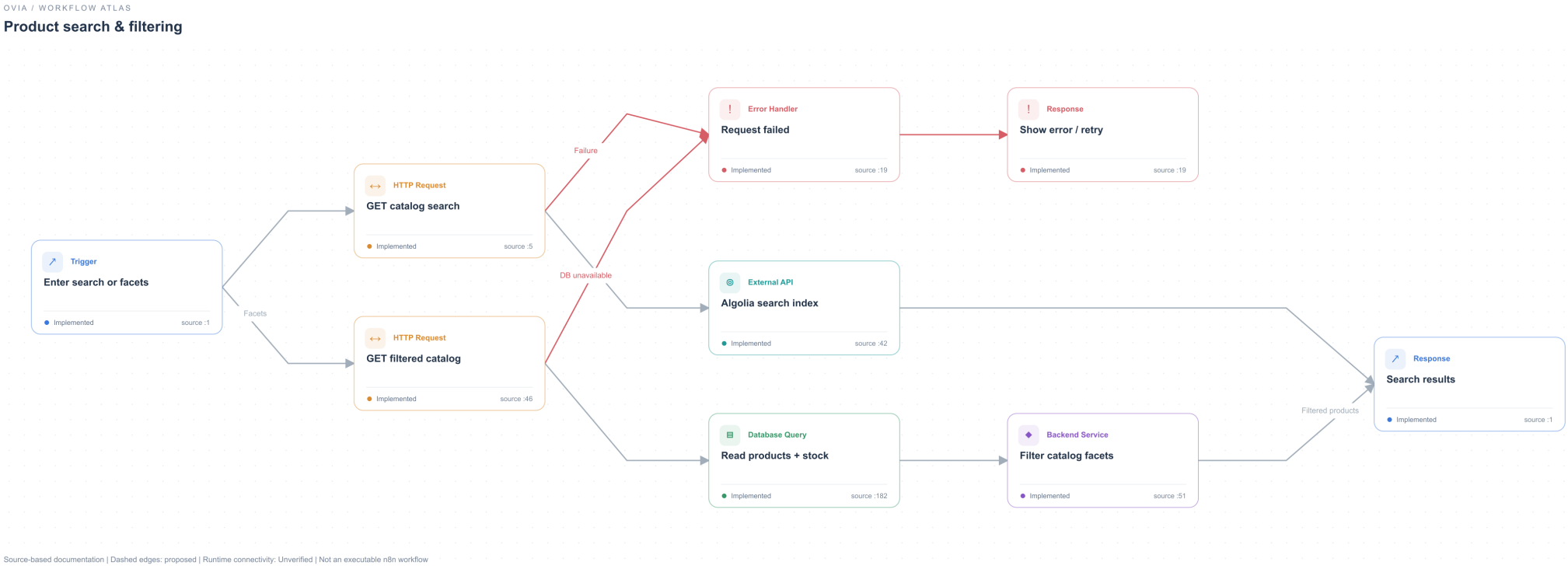
Product search & filtering

Architecture overview. Readable node and connection sheets follow.



Product search & filtering

Readable detail sheet 1/3. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Enter search or facets

Instant search modal and catalog filter UI.

src/components/instant-search-modal.tsx:1

Next -> #2 (p2); Facets -> #7 (p3)

#02 / HTTP Request / Implemented

GET catalog search

Query, limit and page are normalized; the route enforces a ten-second search timeout.

src/app/api/search/route.ts:5

Next -> #3 (p2); Failure -> #5 (p3)

#03 / External API / Implemented

Algolia search index

searchSingleIndex queries the configured products index. Runtime configuration and connectivity are Unverified.

src/lib/algolia.ts:42

Next -> #4 (p2)

#04 / Response / Implemented

Search results

Render returned hits and links to products.

src/components/instant-search-modal.tsx:1

END: no outgoing connection

Product search & filtering

Readable detail sheet 2/3. Node numbers and page references preserve every directed connection.

#05 / Error Handler / Implemented

Request failed

Unavailable configuration, timeout or provider failure returns HTTP 503.

src/app/api/search/route.ts:19

Next -> #6 (p3)

#06 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/app/api/search/route.ts:19

END: no outgoing connection

#07 / HTTP Request / Implemented

GET filtered catalog

Database catalog filtering via filterProducts.

src/app/api/v1/[[...path]]/route.ts:46

Next -> #9 (p4); DB unavailable -> #5 (p3)

#08 / Backend Service / Implemented

Filter catalog facets

Category, department and other Filters constrain product results.

src/lib/catalog-filter.ts:51

Filtered products -> #4 (p2)

Product search & filtering

Readable detail sheet 3/3. Node numbers and page references preserve every directed connection.

#09 / Database Query / Implemented

Read products + stock

Database catalog and SKU inventory balances.

src/server/db.ts:182

Next -> #8 (p3)

Product search & filtering

Complete directed connection register / no omitted success, failure or proposed edges

#01 Enter search or facets -> #02 GET catalog search | Next

#02 GET catalog search -> #03 Algolia search index | Next

#03 Algolia search index -> #04 Search results | Next

#02 GET catalog search -> #05 Request failed | Failure | FAILURE

#05 Request failed -> #06 Show error / retry | Next | FAILURE

#01 Enter search or facets -> #07 GET filtered catalog | Facets

#07 GET filtered catalog -> #09 Read products + stock | Next

#09 Read products + stock -> #08 Filter catalog facets | Next

#08 Filter catalog facets -> #04 Search results | Filtered products

#07 GET filtered catalog -> #05 Request failed | DB unavailable | FAILURE

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Note: GET /api/v1/products and /api/v1/search filter the MongoDB catalog; /api/search uses Algolia.